$\theta \in \mathbb{R}^{n+1}$

$x_0 \equiv 1$

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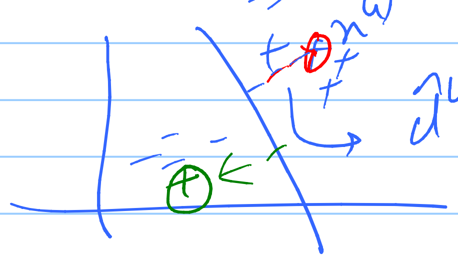
Logistic Regression

$\to \theta^T x \geq 0$

$x^{(i)}$



$z^{(i)} = \theta^T x^{(i)}$

$\{x^{(i)}, y^{(i)}\}_{i=1}^{m}$

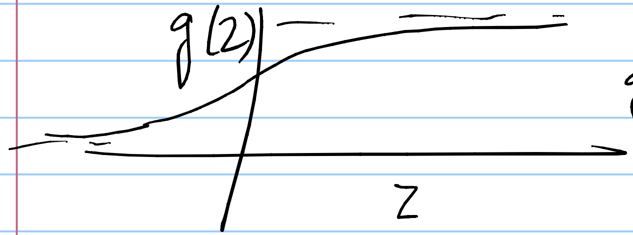
$h_\theta(x) : \mathcal{X} \to \mathcal{Y}$

$\mathcal{X} = \mathbb{R}^n \qquad \mathcal{Y} = \{0, 1\}$

$P(y^{(i)} \mid x^{(i)}; \theta)$

$y^{(i)} \mid x^{(i)}; \theta \sim$

$y \mid x; \theta \sim \text{Bernoulli}(\phi)$



$g(z) = \frac{1}{1+e^{-z}}$

$\sigma(z) = \frac{1}{1+e^{-z}}$ $\to$ Assumption (Modeling)

$\phi^{(i)} = \cancel{\frac{1}{1+e}}\ \sigma(z^{(i)}) = \frac{1}{1+e^{-z^{(i)}}}$

Ri:- $\underline{P(y^{(i)}|x^{(i)};\theta)}$ ✓

Likelihood $L(\theta) = \prod_{i=1}^{m} P(y^{(i)}|x^{(i)};\theta)$ $(x^{(i)}, y^{(i)})$ $\underline{\text{i.i.d.}}$

$$LL(\theta) = \log \prod_{i=1}^{m} P(y^{(i)}|x^{(i)};\theta)$$

$\underline{\text{MLE}}$:- $\arg\max_{\theta} LL(\theta) = \sum_{i=1}^{m} \log \underbrace{P(y^{(i)}|x^{(i)};\theta)}$

Maximum Likelihood $\underline{\text{Estimator}}$

$$\log \left[[\phi^{(i)}]^{y^{(i)}} [1-\phi^{(i)}]^{(1-y^{(i)})} \right]$$

$\theta^{(t+1)} \leftarrow \theta^{(t)} + \underset{\text{Learning Rate}}{\gamma} \cdot \nabla_\theta LL(\theta)$

⇓

Gradient Ascent Update

$$LL(\theta) = \sum_{i=1}^{m} \left[y^{(i)} \log \phi^{(i)} + (1-y^{(i)}) \log [1-\phi^{(i)}] \right]$$

$$\nabla_\theta LL(\theta) = \sum_{i=1}^{m} \left[y^{(i)} \nabla_\theta \log \phi^{(i)} + (1-y^{(i)}) \nabla_\theta \log (1-\phi^{(i)}) \right]$$

$$\phi^{(i)} = \sigma(\theta^T x^{(i)}) = \frac{1}{1+e^{-\theta^T x^{(i)}}}$$

$$= \nabla_\theta \mathcal{L}(\theta) = \sum_{i=1}^{m} \frac{y^{(i)}}{\phi^{(i)}} \nabla_\theta \sigma[\theta^T x^{(i)}] + \frac{[1-y^{(i)}]\,(-1)\,\nabla_\theta \sigma(\theta^T x^{(i)})}{[1-\phi^{(i)}]}$$

$$= \sum_{i=1}^{m} \left[\frac{y^{(i)}}{\phi^{(i)}} + (-1)\frac{(1-y^{(i)})}{(1-\phi^{(i)})} \right] \underbrace{\nabla_\theta \sigma[(\theta^T x^{(i)})]}_{}$$

$$\textcircled{1} \left\{ \frac{\partial\, [\sigma(\theta^T x^{(i)})]}{\partial\, [\theta^T x^{(i)}]} \cdot \nabla_\theta [\theta^T x^{(i)}] \right. \quad \textcircled{2}$$

$$= \sum_{i=1}^{m} \left[\frac{y^{(i)}(1-\phi^{(i)}) - (1-y^{(i)})\,\phi^{(i)}}{\cancel{\phi^{(i)}}\,\cancel{(1-\phi^{(i)})}} \right] \cancel{\phi^{(i)}(1-\phi^{(i)})}\; x^{(i)}$$

Aside :-

$$\sigma'(z) = \sigma(z)\,(1-\sigma(z))$$

Sigmoid Prove $\rightarrow$

$$= \sum_{i=1}^{m} \left[y^{(i)} - \phi^{(i)} \right] x^{(i)}$$

$$= \sum_{i=1}^{m} \left[y^{(i)} - \underbrace{\sigma(\theta^T x^{(i)})}_{} \right] x^{(i)}$$

$\Rightarrow h_\theta(x^{(i)})$

$h_\theta(x) : \sigma(\theta^T x)$ $\qquad h_\theta : X \to (0,1)$

$$\nabla_\theta LL(\theta) = \sum_{i=1}^{m} \left[y^{(i)} - h_\theta(x^{(i)}) \right] x^{(i)}$$

$$\boxed{h_\theta(x^{(i)}) = \sigma(\theta^T (x^{(i)}))}$$ sigmoid fn.

$t \leftarrow 0;$

$\theta^{(t)} \leftarrow init();$

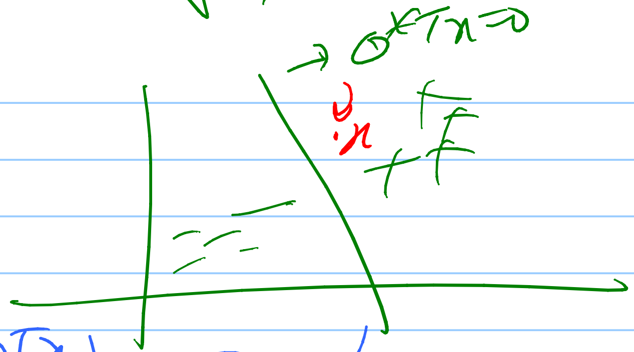
do {

$\theta^{(t+1)} \leftarrow \theta^{(t)} + \eta \cdot \nabla_\theta LL(\theta)$

$t \leftarrow t+1;$

} (while ! converged)

θ^* :- optimal value of parameters



Inference / Test time

$:\rightarrow x$

Compute $h_\theta(x) = \sigma(\theta^T x) = \dfrac{1}{1+e^{-\theta^T x}} = 1/2$

Separation :- $h_\theta(x) = 1/2$

$$\Rightarrow 2 = 1 + e^{-\theta^T x}$$

$$\Rightarrow 1 = e^{-\theta^T x}$$

$$\Rightarrow \boxed{0 = \theta^T x}$$